# Layer 3 Switch Inter-VLAN Routing Lab Guide

*CCNA Hands-on Configuration, Verification & Subnet Mask Refactoring*

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## 1. Theory & Technical Overview

Inter-VLAN routing is the process of forwarding network traffic from one Virtual Local Area Network (VLAN) to another. By default, devices on different VLANs reside in distinct broadcast domains and cannot communicate directly at Layer 2. A Layer 3 device is required to perform IP routing between these VLANs.

### Why Use a Layer 3 Switch for Inter-VLAN Routing?

**ROAS Limitation:** In traditional network designs, Inter-VLAN routing was handled via Router-on-a-Stick (ROAS), where a single trunk link connected a Layer 2 switch to an external router utilizing subinterfaces. While functional for smaller environments, ROAS creates a significant bandwidth bottleneck over the physical trunk link.

**Layer 3 Advantage:** A Layer 3 Switch (Multilayer Switch) combines high-density hardware switching with full IP routing capability inside a single chassis. It utilizes Switch Virtual Interfaces (SVIs)—logical Layer 3 interfaces corresponding to specific VLANs—to route traffic at wire speed using specialized ASIC hardware.

| Feature / Metric | Router-on-a-Stick (ROAS) | Layer 3 Switch (SVI) |
|---|---|---|
| Routing Hardware | Software/CPU-based on external router | Hardware ASIC-driven internal routing |
| Throughput & Latency | Limited by single physical trunk link link speed | Full backplane switching/routing speed |
| Scalability & Setup | Requires router subinterfaces & trunking | Simple internal SVIs (`interface vlan X`) |

## 2. Hands-on Lab Configuration & Execution

### Task 1: VLAN Creation & Port Assignment

Create VLANs 10 (Sales), 20 (Support), and 30 (Marketing) on the Cisco Layer 3 switch (e.g., Cisco WS-C3560 or Cisco Packet Tracer 3560/3650 switch) and assign access ports to each VLAN.

```
L3-Switch# configure terminal
```

```
L3-Switch(config)# vlan 10
L3-Switch(config-vlan)# name Sales
L3-Switch(config-vlan)# vlan 20
L3-Switch(config-vlan)# name Support
L3-Switch(config-vlan)# vlan 30
L3-Switch(config-vlan)# name Marketing
L3-Switch(config-vlan)# exit

! Assign interfaces to respective VLANs
L3-Switch(config)# interface FastEthernet0/1
L3-Switch(config-if)# switchport mode access
L3-Switch(config-if)# switchport access vlan 10
L3-Switch(config-if)# exit

L3-Switch(config)# interface FastEthernet0/2
L3-Switch(config-if)# switchport mode access
L3-Switch(config-if)# switchport access vlan 20
L3-Switch(config-if)# exit

L3-Switch(config)# interface FastEthernet0/3
L3-Switch(config-if)# switchport mode access
L3-Switch(config-if)# switchport access vlan 30
L3-Switch(config-if)# exit
```

## Task 2: Switch Virtual Interfaces (SVI) Configuration

Configure SVIs for each created VLAN to serve as default gateways for connected end devices. Execute 'no shutdown' to ensure SVIs enter the 'up/up' state.

```
L3-Switch(config)# interface vlan 10
L3-Switch(config-if)# ip address 192.168.10.1 255.255.255.0
L3-Switch(config-if)# no shutdown
L3-Switch(config-if)# exit

L3-Switch(config)# interface vlan 20
L3-Switch(config-if)# ip address 192.168.20.1 255.255.255.0
L3-Switch(config-if)# no shutdown
L3-Switch(config-if)# exit

L3-Switch(config)# interface vlan 30
L3-Switch(config-if)# ip address 192.168.30.1 255.255.255.0
L3-Switch(config-if)# no shutdown
L3-Switch(config-if)# exit
```

**SVI Operational Requirement:** An SVI will show Line Protocol DOWN unless: (1) the VLAN exists in the switch VLAN database, and (2) at least one physical port assigned to that VLAN is active (up/up) or a trunk carrying that VLAN is active.

## Task 3: Enabling Global IP Routing

By default, Cisco Layer 3 switches act purely as Layer 2 devices. Global IP routing must be explicitly activated to enable packet forwarding across SVIs.

```
L3-Switch(config)# ip routing
L3-Switch(config)# exit
L3-Switch# show ip route
```

**Verification Step:** After issuing 'ip routing', verify the routing table using 'show ip route'. You will observe Connected (C) and Local (L) routes automatically created for each active SVI subnet.

## Task 4: Inter-VLAN Routing Verification & Ping Logs

Connect PC-1 (VLAN 10) and PC-2 (VLAN 20) with appropriate IP configurations, then perform ICMP ping testing.

| Device Name | VLAN | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC-1 (Sales) | VLAN 10 | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 |
| PC-2 (Support) | VLAN 20 | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 |

### Simulated ICMP Ping Verification Log (PC-1 to PC-2):

```
C:\> ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:
Request timed out.                          <-- ARP Resolution on Gateway
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

**Technical Note:** The initial 'Request timed out' packet is expected behavior on Cisco devices due to Address Resolution Protocol (ARP) resolution occurring at the SVI default gateway for the target subnet on the first ICMP request.

# 3. Task 5: VLAN 30 Refactoring & Subnet Mask Analysis

In Task 5, VLAN 30 (Marketing) is refactored from a /24 prefix (255.255.255.0) to a /25 prefix (255.255.255.128). Below is the configuration refactoring and network impact analysis.

## SVI Refactoring Configuration Commands

```
L3-Switch(config)# interface vlan 30
L3-Switch(config-if)# ip address 192.168.30.1 255.255.255.128
L3-Switch(config-if)# exit

! Verify updated routing table entry
L3-Switch# show ip route connected
C    192.168.30.0/25 is directly connected, VLAN30
```

## Detailed Technical Impact & Analysis

**1. Host Address Space & Capacity:** Changing the subnet mask from 255.255.255.0 (/24) to 255.255.255.128 (/25) divides the 192.168.30.0 network block into two equal subnets of 128 IP addresses each. VLAN 30 now operates strictly within the first subnet range (192.168.30.0 to 192.168.30.127).

• Network ID: 192.168.30.0/25

• Default Gateway (SVI): 192.168.30.1

• Usable Host Range: 192.168.30.1 to 192.168.30.126 (126 total usable host IPs, reduced from 254 usable IPs)

• Directed Broadcast Address: 192.168.30.127

**2. Impact on Inter-VLAN Communication:** Because global 'ip routing' is enabled on the Layer 3 switch, Inter-VLAN communication between VLAN 30 and VLANs 10/20 remains fully functional without requiring changes on VLAN 10 or 20. The switch routes packets based on Longest Prefix Match (LPM) in its routing table.

**3. Host Configuration Requirement:** Any host assigned an IP address between 192.168.30.128 and 192.168.30.254 on VLAN 30 will no longer be in the SVI's local subnet. If a PC is assigned 192.168.30.130/25, it will attempt to reach the gateway 192.168.30.1 on a different subnet, resulting in packet drop. Hosts on VLAN 30 must have their subnet mask updated to 255.255.255.128.